# Prime-Resonant Graph Databases: Non-Local Data Distribution via Prime Hilbert Spaces

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**Abstract**

We present the *Prime-Resonant Graph Database (PR-Graph)*, a distributed data system that encodes vertices, edges, and properties as prime-basis superpositions with phase-encoded payloads. Nodes exchange only compact *resonance beacons* for discovery and epoch coordination, while actual data become reconstructable non-locally via *resonance locking* and *Chinese Remainder* reconstruction. Consistency arises from entropy-guided stabilization, yielding *Resonant Eventual Consistency (REC)*. We formalize the encoding, retrieval, and stability conditions; give a concrete architecture and algorithms; and provide figures illustrating the full pipeline from prime selection to non-local reads and protocol beacons.

## Contents

# 1 Introduction

Conventional distributed databases replicate data by explicit transfer between nodes, incurring bandwidth, coordination, and consistency costs. PR-Graph offers an alternative: represent data as resonance states over a prime-based Hilbert space. Nodes share beacons that describe prime indices, epochs, and coarse fingerprints; payloads remain latent until a reader synthesizes the correct probe to induce a resonance lock. This supports secure, low-bandwidth, and topology-agnostic availability.

**Contributions.** (i) A prime-basis Hilbert encoding for graph atoms with phase-coded payload symbols; (ii) a non-local retrieval process based on resonance overlap and entropy-driven stabilization; (iii) a practical beacon gossip plane and epoch model; (iv) algorithms and complexity analysis; (v) a formal definition of *Resonant Eventual Consistency (REC)*; (vi) protocol-oriented figures including a byte-level beacon layout.

# 2 Prime Hilbert Space Representation

Let $\mathbb{P}$ denote the set of primes and $\{|p\rangle\}_{p\in\mathbb{P}}$ an orthonormal basis. A graph atom $a$ (vertex, edge, or property blob) is addressed by

$$|A(a)\rangle = \sum_{p\in P(a)} c_p\, e^{i\theta_p} |p\rangle, \qquad \sum_{p\in P(a)} |c_p|^2 = 1, \tag{1}$$

where $P(a) \subset \mathbb{P}$ is a finite *prime index* chosen deterministically from a *prime-entropy hash* $H(a)$.

## 2.1 Prime-Entropy Hash and Index Selection

Given a canonical key for $a$ (e.g., a vertex ID, edge tuple, or property key), compute $H(a)$ with a cryptographic hash. Use $H(a)$ to seed a deterministic sampler over a sieve to select $k$ primes:

$$P(a) = \{p_1, \ldots, p_k\}, \quad k \in \{16, 32, 48, 64\}. \tag{2}$$

Weights $c_p$ may be uniform or tapered (e.g., $c_{p_i} \propto p_i^{-\alpha}$ with small $\alpha$) to improve lock robustness.

## 2.2 Phase Coding of Payload Symbols

Payload bytes are chunked into integers $\{m_j\}$. For each chunk $m_j$ and each $p_i \in P(a)$, set

$$\theta_{p_i}^{(j)} = 2\pi \frac{m_j \bmod p_i}{p_i} \; + \; \alpha/\varphi \; + \; \beta/\delta_S \; + \; \mathrm{HMAC}_K(p_i \| j), \tag{3}$$

where $\varphi$ and $\delta_S$ denote golden and silver ratios, and $K$ is an optional secret phase key controlling access.

# 3 Non-Local Retrieval and Stabilization

A reader forms a probe state

$$|Q\rangle = \sum_{p \in P_Q} w_p e^{i\varphi_p} |p\rangle, \tag{4}$$

and evaluates the overlap $R = |\langle Q|A(a)\rangle|$. Retrieval succeeds when $R$ exceeds a threshold *and* an entropy-guided lock condition holds.

## 3.1 Entropy-Guided Resonance Lock

Let $S(t)$ denote symbolic entropy of the local resonance process and $R_S(t)$ a composite resonance score. We model stabilization by

$$\frac{dS}{dt} = -\lambda S, \qquad \lim_{t\to\infty} S(t) \to 0, \qquad R_S(t) \nearrow R_\star. \tag{5}$$

A lock is declared at $(t^*)$ when $S(t^*) < S_{\min}$ and $R_S(t^*) > R_{\min}$ with bounded drift tolerance.

## 3.2 Residue Recovery and CRT

On lock, phase differences $\Delta\phi_{p_i}^{(j)}$ yield residues $m_j \bmod p_i$. Reconstruct $m_j$ via Chinese Remainder Theorem (CRT) with moduli $\{p_i\}$:

$$m_j = \mathrm{CRT}\Big(\{(r_i, p_i)\}_{i=1}^{k}\Big), \quad r_i \equiv m_j \pmod{p_i}, \tag{6}$$

assuming $\prod_i p_i$ exceeds the symbol alphabet range.

**Theorem 1** (Reconstructability)**.** *Let $M$ bound $m_j \in [0, M)$. If $\prod_{i=1}^{k} p_i \geq M$ and residue recovery is correct for at least a co-prime subset whose product $\geq M$, then $m_j$ is uniquely reconstructed by CRT.*

# 4 Graph Model and Hamiltonian Bias

Vertices $v$ use addresses $|A(v)\rangle$ per (1). An edge $e = (u, v)$ is encoded

$$A(e) = R(n)\Big(A(u) \otimes A(v)\Big), \quad n = \mathrm{hash}(u, v, s_e), \tag{7}$$

with a resonance operator $R(n)$ derived from an edge salt $s_e$.

**Subgraph Coherence.** For a subgraph $G = (V, E)$, define

$$H_G = \sum_{(i,j)\in E} J_{ij}\, R_i R_j + \sum_{i\in V} h_i R_i, \tag{8}$$

where $R_i$ are local resonance observables; $H_G$ biases the lock toward coherent cuts, improving traversal queries.

# 5 Consistency: Resonant Eventual Consistency (REC)

Writes create a new epoch $t_e$ with updated phases. Nodes gossip beacons containing $(P(a), t_e, \text{fingerprint}, \text{sig}$ Conflicts resolve by *thermodynamic selection*: the stabilized state with lower steady entropy and higher resonance wins. [REC] A PR-Graph deployment is *resonantly eventually consistent* if, for every atom $a$, all correct observers that continue attempting reads eventually lock to the same stabilized state for $a$.

# 6 System Architecture

## 6.1 Planes

**Gossip plane** (UDP/QUIC): beacons, liveness, epoch skew, Bloom filters of seen indices.
**Resonance plane**: local synthesis of probes, locking dynamics, residue extraction, CRT.

## 6.2 Epochs and Drift

Epoch period $\tau$ (e.g., 13s) bounds phase drift. Readers consider a sliding window of recent epochs and select the candidate maximizing $R_S$ while minimizing stabilized $S$.

# 7 Algorithms

## 7.1 Write

**Algorithm 1** PRG-Put$(\text{key}, \text{payload}, K?)$

$P \leftarrow \textsc{SelectPrimes}(\textsc{Hash}(\text{key}))$
$\{\theta_{p_i}^{(j)}\} \leftarrow \textsc{EncodePhases}(\text{payload}, P, K)$ ▷ Eq. (3)
$\text{fp} \leftarrow \textsc{QuantizeFingerprint}(\{\theta\})$
$t_e \leftarrow \textsc{CurrentEpoch}();\quad \text{sig} \leftarrow \textsc{Sign}(P, t_e, \text{fp})$
$\textsc{GossipBeacon}(P, t_e, \text{fp}, \text{sig})$
**return** $(P, t_e)$

## 7.2 Read

**Algorithm 2** PRG-Get(key, $K$?)

$P \leftarrow \textsc{SelectPrimes}(\textsc{Hash}(\text{key}))$
$\mathcal{B} \leftarrow \textsc{FetchBeacons}(P)$
$(P, t^\star, \text{fp}) \leftarrow \textsc{ChooseCandidate}(\mathcal{B})$ ▷ maximize $R_S$, minimize stabilized $S$
$\Psi_0 \leftarrow \textsc{InitProbe}(P, K)$
$\Psi^* \leftarrow \textsc{ResonanceLock}(\Psi_0, P, t^\star)$ ▷ iterate until $S < S_{\min}$ and $R_S > R_{\min}$
$\{r_i\} \leftarrow \textsc{MeasureResidues}(\Psi^*, P)$
payload $\leftarrow$ $\text{CRT}\textsc{Reconstruct}(\{(r_i, p_i)\})$
**return** payload

## 7.3 Complexity (dominant factors)

Phase synthesis and overlap updates are $O(k)$ per iteration; residue extraction is $O(k)$; CRT with pairwise coprime moduli is $\tilde{O}(k \log^2 M)$ (product tree or Garner). Gossip beacons are $O(k)$ size but highly compressible via delta and index codecs.

# 8 Security and Access Control

Security is enforced by phase knowledge: without $K$, synthesized probes fail to align, keeping $R$ subthreshold. Integrity is verified via beacon signatures and per-chunk MACs embedded into the phase recipe. Optional *semantic cloaking* folds concept keys into phases, restricting reconstruction to holders of topic keys.

# 9 Figures

## Figure 1: System Overview

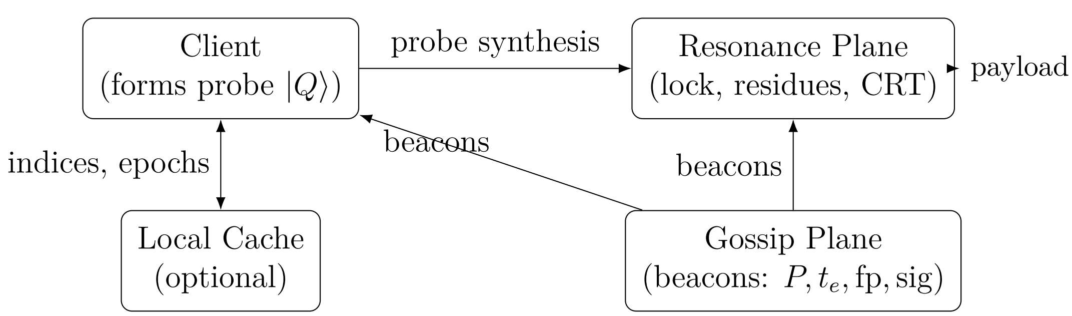


Figure 1: Nodes exchange only beacons via the gossip plane; payloads are reconstructed locally after a resonance lock on the resonance plane.

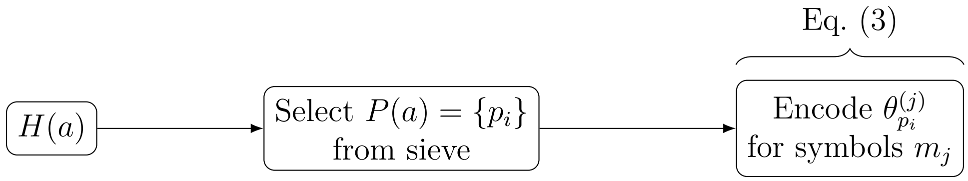


Figure 2: Deterministic prime index selection followed by symbol-wise phase encoding.

## Figure 2: Prime Selection and Phase Encoding

## Figure 3: Resonance Lock Dynamics

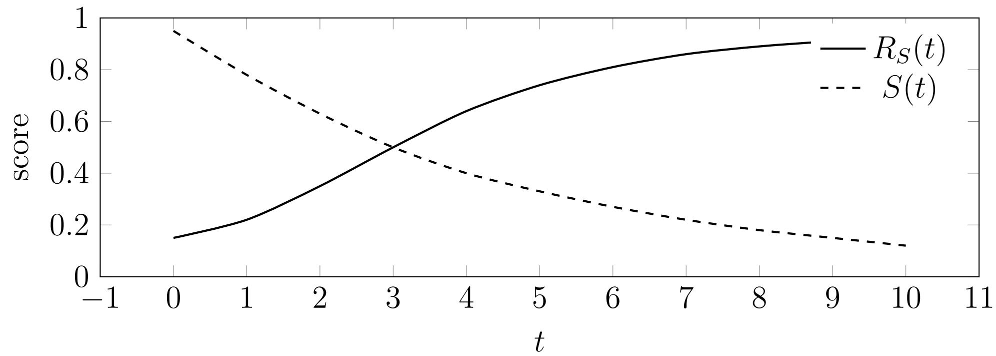


Figure 3: Typical evolution toward lock: resonance score $R_S$ rises as symbolic entropy $S$ decays.

## Figure 4: CRT Reconstruction Pipeline

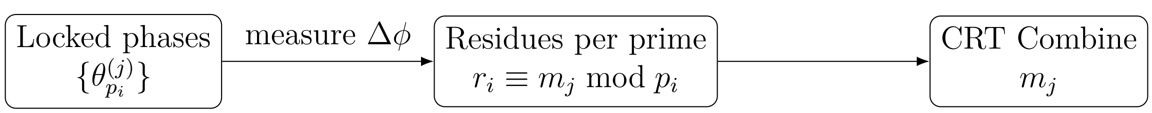


Figure 4: From locked phases to residues to symbol reconstruction with CRT.

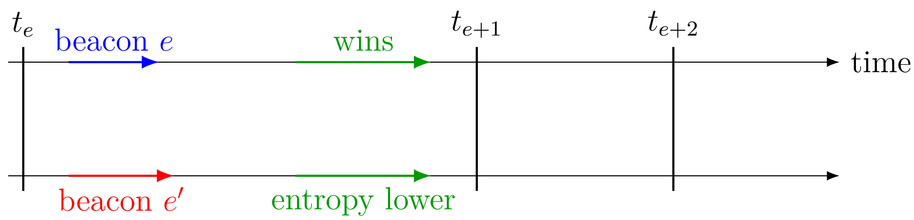


Figure 5: Competing writes $e$ and $e'$ converge by selecting the lower-entropy stabilized state (REC).

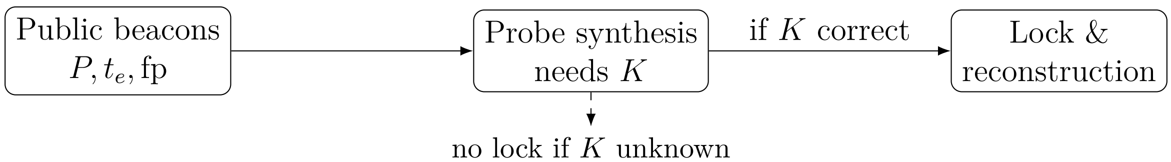


Figure 6: Beacons reveal indices, not payload. Only holders of phase key $K$ can synthesize locking probes.

### Figure 5: REC Timeline and Conflict Resolution

### Figure 6: Security by Phase Knowledge

### Figure 7: Beacon Packet Layout (Conceptual)

### Figure 8: Prime-Index Codec Schematic

### Figure 9: Beacon Byte-Level Layout (RFC-style)

## 10 Implementation Notes

**Library.** Core functions: `selectPrimes`, `encodePhases`, `lockStep` (iterative resonance), `measureResidues`, `crtReconstruct`.

**Gossip.** QUIC/UDP with signed beacons and small Bloom filters of $P(a)$.

**Parameters.** Typical $k = 32$ primes; epoch $\tau = 2$s; thresholds $(S_{\min}, R_{\min})$ tuned empirically.

## 11 Discussion

PR-Graph reframes storage as a resonance schema. Data are everywhere potentially present and become concretely available when a qualified observer locks onto the correct prime-phase pattern. This reduces bandwidth, supports graceful partition tolerance, and enables access control through phase knowledge.

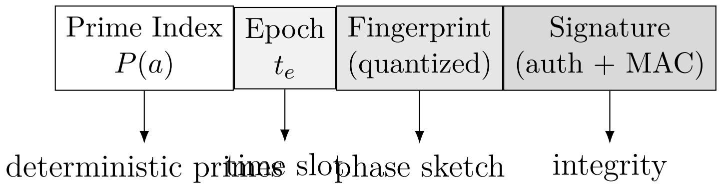


Figure 7: Beacon packets exchanged in the gossip plane: compact metadata that allows synchronization without carrying payloads.

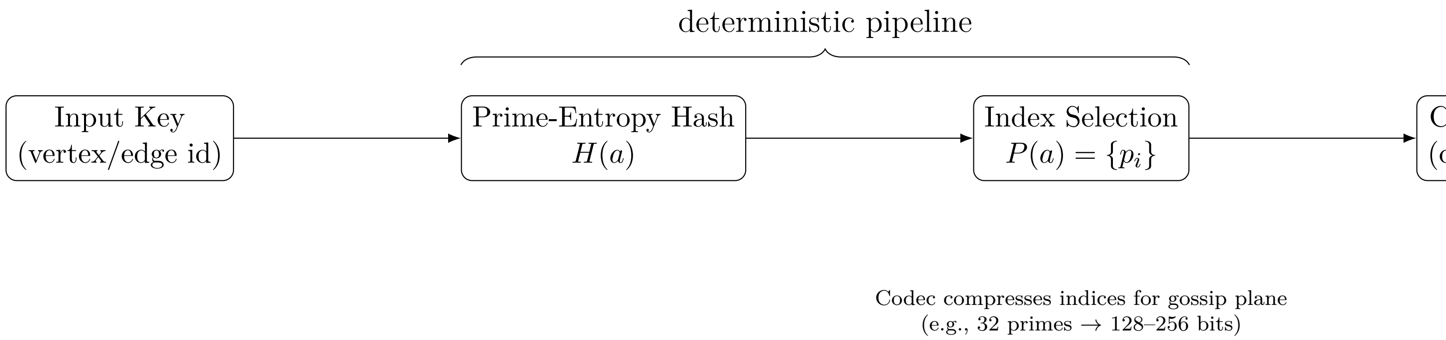


Figure 8: Prime-index codec pipeline: from input key to prime entropy hash, index selection, and compressed beacon representation.

# 12 Conclusion

We developed a full formalism and architecture for prime-resonant, non-local data distribution in graph databases, together with stabilization dynamics and a practical beaconing layer. The approach invites new applications in secure synchronization, semantic access control, and resonance-based computation.


# Acknowledgments

We acknowledge ongoing work integrating entropy spectrometry, holographic quantum encoding, and non-local communication within this framework.


# References


[1] S. Schepis, *Entropy Spectrometry and the Measurement of Awareness*, 2025.

[2] S. Schepis, *Abstract Quantum Mechanics: Non-Local Communication Through Synchronized Mathematical Structures*, 2025.

[3] S. Schepis, *Unified Physics from Consciousness-Based Resonance*, 2025.

| Bytes 0–15 | 16–19 | 20–35 | 36–67 |
|---|---|---|---|
| Prime Index ID (32–128 bits) | Epoch (32 bits) | Phase Fingerprint (64–128 bits) | Signature / MAC (256 bits) |

Figure 9: Byte-level beacon layout, RFC-style. The prime index ID encodes selected primes (delta-coded or Bloom-compressed), epoch encodes time slot, fingerprint provides coarse phase quantization, and signature authenticates the packet.